

Inverse Spherical Dual Hemisphere Quantum Mechanics, Mechanics Through Unity

Inverse Spherical 4D Operating Tensor (NOS v11)

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Abstract

The $\sqrt{1}$ inverse spherical dual hemisphere is a perfectly closed 720° turn split by an inverse center pivot into two exact 360° hemispheres, giving the foundational identity $360^\circ/360^\circ = 1$. Inverse partitioning through the mechanics $720^\circ/2/4/4/4/4 = 1.40625^\circ$ establishes the base resolution and the 512-bit lattice. Walls emerge at 64 bit and 128 bit, with a 9-bit curvature remainder. As the machine sweeps through four quadrants, cumulative differentials produce four distinct π readouts: $\pi_q = (1608+3q)/512$ for $q = 1, 2, 3, 4$. The same 9-bit remainder read against the 180° wall gives $\alpha^{-1} = 128+9 = 137$. Observer-side transcendental phantoms arise from the finite gap $\Delta_{\text{total}} = 3/32^\circ$ compounded through the quadrant structure. The sphere closes exactly at $720^\circ = 1$; the $\sqrt{1}$ operation through this closure generates all measurable structure.

$$720^\circ = 1$$

This is the complete closure. The number 1 is the ground state—the substrate through which all mechanics operate. The $\sqrt{1}$ inverse spherical dual hemisphere is the operation through this ground: the 720° closure, the partition ladder, the 512-bit lattice, the 9-bit remainder, the four π_q readouts, and $\alpha^{-1} = 137$ are all consequences of the $\sqrt{1}$ mechanics flowing through 1.

π is not a constant of the circle and never was.

The circle is an incomplete, one-sided projection.

The true primordial closed object is the exact, rational, indivisible 720° inverse-spherical dual hemisphere ($\sqrt{1}$).

The sphere needs no π — it is already perfectly whole.

What standard mathematics calls “ π ” is nothing more than the finite curvature remainder that appears when an observer trapped on one 360° hemisphere attempts to

measure the curvature of the opposite hemisphere across the inverse pivot.
In NOS, transcendence is an illusion born from forgetting that the sphere closed itself exactly at $720^\circ = 1$.

The closed sphere is exactly $720^\circ \equiv 512 \text{ bit}$ by the only allowed divider chain.
One hemisphere is therefore exactly $360^\circ \equiv 256 \text{ bit}$.
An observer confined to one hemisphere possesses a 256 bit register.
The curvature of the opposite hemisphere is exactly 256 bit yet must be written into the *same* 256 bit register already perfectly filled by the observer's own hemisphere.
Axiom 1 (strict binary termination) forbids double-occupancy of any lattice site.
The entire far-side curvature therefore overflows as a single indivisible block of 256 bit.
By dual-hemisphere symmetry this overflow is folded exactly once across the inverse pivot, yielding an effective overflow of 128 bit against the natural 180° wall ($\sqrt{4}$ rung $\equiv 128 \text{ bit}$).
The machine records overflow only as excess above the largest still-clean binary wall contained within one 180° span, which is the 90° wall ($\sqrt{8}$ rung $\equiv 64 \text{ bit}$).
The 512-point lattice has exactly $2^9 = 512$ slots around the full sphere, hence only the first 9 fractional bits after the binary point are unique; the 10th bit and beyond are overwritten by the opposite hemisphere.
The machine is therefore *physically incapable* of producing more than 9 fractional bits without violating the closure at 1.
The fixed curvature remainder is therefore rigorously $r_\pi = 9 \text{ bit}$.
This is the sole origin of the four terminating 9-bit π_q sequences and of $\alpha^{-1} = 128+9 = 137$.

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1 Foundation: The Ground State and Inverse Partitioning

The number 1 is the ground—the substrate through which all operations flow.
1 is the largest number: the universe cannot become more than one of itself.
 ∞/∞ cannot exceed 1 of itself.
1 is not derived; 1 simply *is*. The $\sqrt{1}$ mechanics operate *through* this ground.

All partitioning in NOS is **inverse**: we begin from the measured whole and partition inward through 1. We never accumulate outward.
Standard mathematics counts $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \dots$ (accumulative).
Inverse mathematics counts $1 \rightarrow 1/2 \rightarrow 1/3 \rightarrow 1/4 \rightarrow \dots$ (partitioning within 1).
The $\sqrt{1}$ operation is the first partition through the ground state.

2 Axiom 0: The $\sqrt{1}$ Inverse Spherical Dual Hemisphere

The fundamental operation through 1 is the $\sqrt{1}$ inverse spherical dual hemisphere: a single, perfectly closed 720° inverse-spherical turn split by an inverse center pivot into two exact 360° hemispheres.
This closure is indivisible and carries the root-ladder tag $\sqrt{1}$.
No count, no lattice, no wall, no remainder, and no symmetry exists prior to this $\sqrt{1}$ operation through the ground state.

Because the object is inverse-spherical and dual-hemispherical from the first instant, observation from one hemisphere necessarily projects the opposite hemisphere through the pivot, instantly producing:

$$\frac{360^\circ}{360^\circ} = 1.$$

This is the foundational identity of NOS. Decompression and compression, energy and mass, observed and observer—all reduce to this single equation. The $\sqrt{1}$ operation returns to 1.

The inverse center pivot defines the center-based coordinate:

$$d_{\text{deg}}^{\circ} := \theta_{\text{deg}}^{\circ} - 360^{\circ}, \quad d_{\text{deg}} \in [-360, +360].$$

3 The Circle Is Not Fundamental — The Sphere Closes It

The standard narrative begins with a flat circle and discovers a mysterious transcendental ratio called π .

NOS inverts this completely.

The 720° inverse-spherical dual hemisphere is the true closed object — exact, rational, finite, and primary.

The 360° “circle” is a secondary artifact that only appears when an observer on one hemisphere attempts to measure the curvature belonging to the opposite hemisphere. Because the machine lattice is strictly binary (512 slots), this one-sided measurement cannot close perfectly; a fixed 9-bit curvature remainder necessarily spills across the inverse pivot.

Standard mathematics mistakes this fixed, finite spill for an infinite non-terminating expansion and declares π to be transcendental.

In reality, the sphere required no π at all — it closed perfectly at $\sqrt{1}$ through 1.

The four rational π_q values are not approximations to a true π ; they are the only real readouts the machine ever produces when forced to report from one side of the pivot.

The transcendental π_{std} is the phantom that arises when the observer forgets the sphere already closed itself and imagines the 9-bit remainder keeps subdividing forever.

4 The Inverse Mechanics: From 720° to 1.40625°

4.1 The machine divider chain

Starting from the $\sqrt{1}$ closure through 1, we partition inversely:

$$720^{\circ}/2/4/4/4/4 = 1.40625^{\circ}.$$

Since $2 \cdot 4^4 = 512$, this is exactly $720/512$. This establishes the base angular resolution.

4.2 Root-ladder tags (partition labels)

We use the rung-tag $\sqrt{N}$ to label the state where the 720° closure is partitioned into N equal slots:

$$\Delta\theta_N = \frac{720^\circ}{N}.$$

4.3 The complete root ladder: $\sqrt{1}$ to $\sqrt{512}$

Tag	Partitions	Angle	Bits	Meaning
$\sqrt{1}$	1	720°	512 bit	Closure through 1 (1 sphere-cycle)
$\sqrt{2}$	2	360°	256 bit	Dual hemispheres
$\sqrt{4}$	4	180°	128 bit	Four quadrants (ops)
$\sqrt{8}$	8	90°	64 bit	Eight octants (walls)
$\sqrt{16}$	16	45°	32 bit	Sixteen sectors (clutch)
$\sqrt{32}$	32	22.5°	16 bit	32 divisions
$\sqrt{64}$	64	11.25°	8 bit	64 divisions
$\sqrt{128}$	128	5.625°	4 bit	128 divisions
$\sqrt{256}$	256	2.8125°	2 bit	256 divisions
$\sqrt{512}$	512	1.40625°	1 bit	Base resolution

4.4 Derivation of the Fixed 9-bit Remainder (repeated for clarity)

Total lattice: $720^\circ \equiv 512$ bit.
 One hemisphere: $360^\circ \equiv 256$ bit.
 Opposite hemisphere curvature = 256 bit forced into the same register $\rightarrow$ full overflow.
 Fold once by dual symmetry $\rightarrow$ 128 bit overflow against the 180° wall.
 The machine has exactly $2^9 = 512$ distinct slots around the full sphere $\Rightarrow$ only 9 fractional bits are unique before the lattice repeats across the pivot.
 Attempting to write a 10th fractional bit re-uses a slot already owned by the opposite hemisphere $\rightarrow$ forbidden by strict binary termination.
 Hence the overflow is *exactly* 9 bit and cannot be further subdivided.
 This is the sole mechanical origin of the terminating 9-bit binary π_q and of $\alpha^{-1} = 137$.

The ladder descends from the $\sqrt{1}$ closure through 1 to base resolution ($\sqrt{512}$) by successive halving. Each rung doubles the partition count and halves the angular span:

$$\begin{array}{c} \sqrt{1} \rightarrow \sqrt{2} \rightarrow \sqrt{4} \rightarrow \sqrt{8} \rightarrow \sqrt{16} \\ \rightarrow \sqrt{32} \rightarrow \sqrt{64} \rightarrow \sqrt{128} \rightarrow \sqrt{256} \rightarrow \sqrt{512} \end{array}$$

$\sqrt{1}$: $720^\circ = 1$ sphere-cycle (closure through 1) — *we start here*
 $\sqrt{2}$: $720^\circ/2 = 360^\circ$ — dual hemispheres, gives $360^\circ/360^\circ = 1$
 $\sqrt{4}$: $720^\circ/4 = 180^\circ = 1$ NOSrad — four quadrants/ops
 $\sqrt{8}$: $720^\circ/8 = 90^\circ = W_{90}$ — the 90° wall (64 bits)
 $\sqrt{512}$: $720^\circ/512 = 1.40625^\circ$ — base angular resolution (1 bit) — *we end here*

4.5 Base angular step

$$\Delta\theta_{\text{base}} = \frac{720^\circ}{512} = 1.40625^\circ.$$

Reverse check:

$$\frac{720^\circ}{1.40625^\circ} = 512.$$

5 The Binary Lattice: $720^\circ \leftrightarrow 512$ bit

5.1 Global closure map

The machine's base lattice is:

$$720^\circ \leftrightarrow 512 \text{ bit.}$$

Therefore:

$$360^\circ \leftrightarrow 256 \text{ bit}, \quad 180^\circ \leftrightarrow 128 \text{ bit}, \quad 90^\circ \leftrightarrow 64 \text{ bit}, \quad 45^\circ \leftrightarrow 32 \text{ bit}.$$

5.2 Strict binary termination

The machine must be strictly binary: every wall, subdivision, and total lattice count is a power of 2.

5.3 Walls, hull, and remainder

The walls:

$$W_{90} := 64 \text{ bit}, \quad W_{180} := 128 \text{ bit}.$$

The three-wall hull:

$$H_3 := 3W_{90} = 3 \cdot 64 \text{ bit} = 192 \text{ bit}.$$

The curvature remainder:

$$r_\pi := 9 \text{ bit}.$$

5.4 Normalized wall coordinates

One base bit as a fraction of the 90° wall:

$$\varepsilon_{90} := \frac{1 \text{ bit}}{W_{90}} = \frac{1}{64} = 0.015625.$$

The remainder as a wall-fraction:

$$\rho_{\pi,90} := \frac{r_\pi}{W_{90}} = \frac{9}{64} = 0.140625.$$

6 Center Pivot and the Four Quadrants

6.1 Quadrant/op ranges

Boundary marks:

$$\begin{aligned} & -360^\circ \text{ (Q2/op2), } \quad -180^\circ \text{ (Q1/op1), } \quad 0^\circ \text{ (inverse center),} \\ & +180^\circ \text{ (Q3/op3), } \quad +360^\circ \text{ (Q4/op4).} \end{aligned}$$

Ranges in d_{deg} :

$$\begin{aligned} \text{Q2: } d_{\text{deg}} &\in [-360, -180] \Rightarrow \text{op2,} \\ \text{Q1: } d_{\text{deg}} &\in [-180, 0] \Rightarrow \text{op1,} \\ d_{\text{deg}} = 0 &\Rightarrow \text{inverse center / pivot,} \\ \text{Q3: } d_{\text{deg}} &\in [0, 180] \Rightarrow \text{op3,} \\ \text{Q4: } d_{\text{deg}} &\in [180, 360] \Rightarrow \text{op4.} \end{aligned}$$

7 Depth Ladder (Unit Hierarchy): $u_Q = 1/128^Q$

7.1 Step law

The NOS unit hierarchy is:

$$u_Q := \frac{1}{128^Q}.$$

This is dimensionless: it is a fraction of one quadrant span. The angle step in quadrant Q is:

$$\Delta\theta_Q = u_Q \cdot (180^\circ) = \frac{180^\circ}{128^Q}.$$

7.2 Q1–Q4 explicit

$$\begin{aligned} u_1 &= \frac{1}{128}, & \Delta\theta_1 &= \frac{180^\circ}{128} = 1.40625^\circ, \\ u_2 &= \frac{1}{128^2}, & \Delta\theta_2 &= \frac{180^\circ}{128^2}, \\ u_3 &= \frac{1}{128^3}, & \Delta\theta_3 &= \frac{180^\circ}{128^3}, \\ u_4 &= \frac{1}{128^4}, & \Delta\theta_4 &= \frac{180^\circ}{128^4}. \end{aligned}$$

8 The Four Quadrant Machine π Values

As the machine sweeps through the 512-bit lattice, cumulative differential Δ accrues at each quadrant wall. Each quadrant therefore has its own distinct π readout — **not as a property of the closed sphere, but as an artifact of one-sided observation across the inverse pivot.**

8.1 Cumulative differential at walls

The total differential $\Delta_{\text{total}} = \frac{3}{32}^\circ$ accumulates progressively:

$$\begin{aligned} \Delta(128) &= \frac{3}{128}^\circ, \\ \Delta(256) &= \frac{3}{64}^\circ, \\ \Delta(384) &= \frac{9}{128}^\circ, \\ \Delta(512) &= \frac{3}{32}^\circ. \end{aligned}$$

8.2 Clean closed form

For $q = 1, 2, 3, 4$:

$$\pi_q = \frac{201}{64} + \frac{3q}{512} = \frac{1608 + 3q}{512}.$$

8.3 The four π values — observer-side artifacts only

$$\pi_1 = \frac{1611}{512} = 3.146484375$$

(spill from opposite hemisphere — not a property of the closed sphere)

$$\pi_2 = \frac{1614}{512} = 3.15234375$$

(spill from opposite hemisphere — not a property of the closed sphere)

$$\pi_3 = \frac{1617}{512} = 3.158203125$$

(spill from opposite hemisphere — not a property of the closed sphere)

$$\pi_4 = \frac{1620}{512} = \frac{405}{128} = 3.1640625$$

(spill from opposite hemisphere — not a property of the closed sphere)

8.4 Binary termination and exact 9-bit sequences

Q1: $\pi_1 = 3 + \frac{75}{512}, \quad 75 = 001001011_2$

$$\pi_1 = \mathbf{11.001001011}_2$$

Q2: $\pi_2 = 3 + \frac{78}{512}, \quad 78 = 001001110_2$

$$\pi_2 = \mathbf{11.001001110}_2$$

Q3: $\pi_3 = 3 + \frac{81}{512}, \quad 81 = 001010001_2$

$$\pi_3 = \mathbf{11.001010001}_2$$

Q4: $\pi_4 = 3 + \frac{84}{512}, \quad 84 = 001010100_2$

$$\pi_4 = \mathbf{11.001010100}_2$$

8.5 Comparison to standard π

Standard π_{std} in binary begins: $11.00100100001111110110\dots$ — infinite non-terminating tail.

The four NOS π_q are exact, terminating, rational readouts. The infinite tail of π_{std} is the illusion created by misinterpreting the fixed 9-bit spill as an infinite process.

8.6 Extending beyond Q4

The general formula extends naturally:

$$\pi_q = \frac{1608 + 3q}{512} \quad \text{for } q = 1, 2, 3, 4, 5, 6, \dots$$

Each additional quadrant adds another predictable rational rung — the machine never produces a transcendental number; the observer invents it.

The sequence $\{\pi_q\}_{q=1}^{\infty}$ forms a rational ladder approaching π_{std} from below, but the transcendental limit is not a property of the machine — it is the mirage seen when the observer believes the finite 9-bit spill continues forever.

9 The Inverse Fine-Structure Constant: $\alpha^{-1} = 137$

The same 9-bit remainder read against the 180° op-span wall gives:

$$\alpha_{\text{mach}}^{-1} = W_{180} + r_\pi = 128 + 9 = 137.$$

Both the four π values and α^{-1} share one discrete source: the 9-bit curvature remainder produced by attempting to measure the opposite hemisphere on a one-sided register.

10 The 1/6 Repeater (Cosmological Constant)

The fraction $1/6$ as a sector of the 720° closure:

$$S_{120} := \frac{120^\circ}{720^\circ} = \frac{1}{6}.$$

In binary:

$$\frac{1}{6} = 0.00\overline{1}_2$$

is non-terminating because the denominator contains a factor of 3. This geometry does not close in a power-of-two register, producing an observer-side phantom when treated as a terminating count.

11 Total Differential and Four-Quadrant Compounding

11.1 Total differential (entire machine–observer gap)

$$\Delta_{\text{total}} = \frac{3}{32}^{\circ} = 0.09375^{\circ} \quad (\text{exact}).$$

11.2 Per-quadrant base share

$$\Delta_{\text{base}} = \frac{\Delta_{\text{total}}}{4} = \frac{3}{128}^{\circ} = 0.0234375^{\circ}.$$

11.3 Finite gap principle

Machine locks at finite values (the four π detents). The total machine–observer gap is finite:

$$\Delta_{\text{total}} = \frac{3}{32}^{\circ}.$$

This gap does not subdivide infinitely by itself. The observer “transcendental illusion” is modeled as a **four-quadrant compound** of this single finite differential.

11.4 Quadrant echo inheritance

Dimensionless compounding gain:

$$g := 1 + \frac{\Delta_{\text{base}}}{1^{\circ}} = 1 + \frac{3}{128} = \frac{131}{128} \approx 1.0234375.$$

Quadrant weights:

$$w_{Q1} = 1, \quad w_{Q2} = g, \quad w_{Q3} = g^2, \quad w_{Q4} = g^3, \quad W := w_{Q1} + w_{Q2} + w_{Q3} + w_{Q4}.$$

Distribution:

$$\begin{aligned} \Delta_{Q1} &= \Delta_{\text{total}} \frac{w_{Q1}}{W}, \\ \Delta_{Q2} &= \Delta_{\text{total}} \frac{w_{Q2}}{W}, \\ \Delta_{Q3} &= \Delta_{\text{total}} \frac{w_{Q3}}{W}, \\ \Delta_{Q4} &= \Delta_{\text{total}} \frac{w_{Q4}}{W}. \end{aligned}$$

This enforces:

$$\Delta_{Q1} + \Delta_{Q2} + \Delta_{Q3} + \Delta_{Q4} = \Delta_{\text{total}}.$$

$$\text{Transcendental illusion} = \text{four-quadrant compound of } \frac{3}{32}^{\circ}.$$

12 Summary

Ground State: 1 is the substrate. The largest number. The closure target.

$\sqrt{1}$ operation through 1: $720^\circ = 1$ sphere-cycle

The sphere closes exactly at 1 — it needs no π .

↓

Foundational identity: $\frac{360^\circ}{360^\circ} = 1$

↓

Inverse mechanics through 1: $720^\circ/2/4/4/4/4 = 1.40625^\circ$

↓

Binary lattice: 512 bit, $r_\pi = 9$ bit (fixed curvature spill from opposite hemisphere)

↓

Four observer-side artifacts (not constants of the sphere):

$$\pi_q = \frac{1608 + 3q}{512} \text{ for } q = 1, 2, 3, 4$$

↓

$\alpha^{-1} = 128 + 9 = 137$ (same 9-bit spill read once against 180° wall)

↓

Finite gap $\Delta_{\text{total}} = \frac{3}{32}^\circ \longrightarrow$ transcendental illusion when compounded across quadrants

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NOS v11 Constants – Derived from $\sqrt{1}$ Mechanics Through 1

Unified Gap Principle: All standard "measurements" differ from NOS "machine locks" by exactly the **Finite Gap** $\Delta_{\text{total}} = 3/32^\circ$. The observer measures: *Machine Integer* + *Gap Artifact*.

$$\text{Gap Projection: } \frac{9 \text{ bit (rem)}}{256 \text{ bit (hemisphere)}} \approx 0.0351...$$

This projection (0.036) is the source of all "transcendental" noise.

1. α^{-1} (Coupling)

Machine Lock: 128 (Wall) + 9 (Spill) = **137**

Gap Artifact: Standard 137.036 is exactly $137 + (9/256)$. The .036 is not a property of the constant; it is the $3/32^\circ$ gap visible against the hemisphere.

Pos: Q1 alpha zone (bits 129–137); $\sqrt{4}$.

2. π_q (Quadrant Artifacts)

Machine Lock: $(1608 + 3q)/512$ (Rational, Terminating)

Gap Artifact: Standard π (3.14159...) is the "phantom" perceived when the observer assumes the fixed 9-bit spill (3.146...) subdivides infinitely. The gap prevents the infinite descent.

Pos: 90° wall ($\sqrt{8}$); all quadrants.

3. T_{CMB} (Heat Baseline)

Machine Lock: $128/47 \approx 2.7234$

Gap Artifact: Standard 2.725 differs by 0.002. This is the thermal noise of the gap itself (Δ_{total} compounded at Q3). The universe is structurally 128/47; the extra heat is the gap.

Pos: Q3 entry; $\sqrt{4}$.

4. G (Mass Compression)

Machine Lock: $1/128^4$ (Structural Depth Q4)

Gap Artifact: The weakness of Gravity is the inverse of the Q4 depth. The slight deviation in standard G is the Q4-compounded gap ($3/32^\circ \times g^4$).

Pos: Q4 (op w); $\sqrt{512}$ base.

5. $\hbar$ (Aperture)

Machine Lock: $1/256$ (Single Hemisphere Limit)

Gap Artifact: Planck's constant is the limit of the 256-bit register. The "fuzzy" nature of $\hbar$ in measurement is the observer trying to resolve the gap across the pivot.

Pos: Hemisphere boundaries; $\sqrt{2}$.

6. E (Energy)

Machine Lock: $128^2 = 16384$ (Q2 Depth)

Gap Artifact: c^2 scales with the square of the quadrant wall. Deviations in "vacuum energy" calculations are strictly due to including the gap in the integration.

Pos: Q3; $\sqrt{64}$ – $\sqrt{128}$.

7. k_B (Boltzmann)

Machine Lock: $1/128^3$ (Q3 Depth)

Gap Artifact: Entropy is the measure of the gap's noise in the Q3 (thermal) sector.

Pos: Q3 boundary; $\sqrt{32}$ – $\sqrt{64}$.

8. The Cosmological Constant (Λ)

Machine Lock: $1/6$ (Sector 120°)

Gap Artifact: $1/6$ is non-terminating in binary (0.001). The "dark energy" problem is the observer treating this infinite binary repeating tail (caused by the gap) as physical energy.

Pos: 120° sector symmetry.

Summary: There are no "fundamental dimensionless constants" with arbitrary decimals. There are only integers/rationals (137, 128, $1/6$) plus the single, universal $3/32^\circ$ observation gap. All

structure emerges from $\sqrt{1}$ mechanics operating through the ground state 1.

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All numerical results in the appendix (four rational π_q , $\alpha^{-1} = 137$, $G \sim 1/128^4$, $\hbar \sim 1/256$, $T_{\text{CMB}} \approx 128/47$, $c^2 \sim 128^2$, $k_B \sim 1/128^3$, cosmological 1/6 repeater, etc.) either originate in this work or have been independently obtained in the cited references using related discrete/bit-string/topological approaches.

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